

MACHINE LEARNING

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5AI19	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. Ability to comprehend the concept of supervised and unsupervised learning techniques
2. Differentiate regression, classification and clustering techniques and to implement their algorithms.
3. analyze the performance of various machine learning techniques and to select appropriate features for training machine learning algorithms.
4. Be familiar with basic machine learning algorithms with classification and clustering.
5. Gain experience of doing independent study and research.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Recognize the characteristics of machine learning that makes it useful to solve real-world problems.
2. Evaluate exploratory data analysis and Data preparation and preprocessing on different datasets.
3. Design and implement machine learning solutions of classification, regression problems.
4. Choose an appropriate clustering technique to solve real world problems.
5. Choose a suitable machine learning model, implement and examine the performance of the chosen model for a given real world problems.

UNIT-I	INTRODUCTION	CLASSES: 12
Machine Learning Foundations: Introduction to machine learning, learning problems and scenarios, need for machine learning, types of learning, standard learning tasks, the Statistical Learning Framework, Probably Approximately Correct (PAC) learning.		
UNIT-II	DESCRIPTIVE STATISTICS	CLASSES: 12
Data representation, types of data- nominal, ordinal, interval and continuous, central tendency- calculating mean mode median, mean vs median, variability, variance, standard deviation, Mean Absolute Deviation using sample dataset, finding the percentile, interquartile range, Box Plot, Outlier, whisker, calculating correlation, covariance, causation. Exploratory data analysis, Data preparation and preprocessing, Data visualization		
UNIT-III	SUPERVISED LEARNING	CLASSES: 14
Learning a Class from Examples, Linear, Non-linear, Multi-class and Multi-label classification, Perceptron: – multilayer neural networks – back propagation - learning neural networks structures – support vector machines: – soft margin SVM – going beyond linearity – generalization and over fitting – regularization – validation. Decision Trees: ID3, Classification and Regression Trees (CART), Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.		
UNIT-IV	UNSUPERVISED LEARNING	CLASSES: 12
Introduction to clustering, Hierarchical: AGNES, DIANA, Partitional: K-means clustering, K-Mode Clustering, Self-Organizing Map, Expectation Maximization, Gaussian Mixture Models, Principal Component Analysis (PCA), Locally Linear Embedding (LLE), Factor Analysis, Neural Networks: Introduction, Perceptron, Multilayer Perceptron, Support vector machines: Linear and NonLinear, Kernel Functions, K-Nearest Neighbors.		

UNIT- V	ENSEMBLE AND PROBABILISTIC LEARNING	CLASSES: 12
Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Gradient Boosting, Xg boost, Stacking Bayesian Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesian Belief Networks, Mining Frequent Patterns		
TEXT BOOKS		
1. Giuseppe Bonaccorso, —Machine Learning AlgorithmsII, 2nd Edition, Packt, 2018, 2. Tom Mitchel —Machine LearningII, Tata McGraw Hill, 2017.		
REFERENCE BOOKS		
1. Ethem Alpaydin, —Introduction to Machine LearningII, PHI, 2004 2. Stephen Marshland, —Machine Learning: An Algorithmic PerspectiveII, CRC Press Taylor & Francis, 2nd Edition, 2015 3. Abhishek Vijavargia —Machine Learning using PythonII, BPB Publications, 1 st Edition, 2018 INTRODUCTION TO MACHINE LEARNING		